

Works both ways

It does not matter which order numbers appear in a multiplication sum because the answer will be the same either way. Two methods of the same multiplication are shown here.

$4 \times 3 = 3 + 3 + 3 + 3 = 12$



$3 \times 4 = 4 + 4 + 4 = 12$



Multiplying by 10, 100, 1,000

Multiplying whole numbers by 10, 100, 1,000, and so on involves adding one zero (0), two zeroes (00), three zeroes (000), and so on to the right of the number being multiplied.

add 0 to end of number →
 $34 \times 10 = 340$

add 00 to end of number →
 $72 \times 100 = 7,200$

add 000 to end of number →
 $18 \times 1,000 = 18,000$

Patterns of multiplication

There are quick ways to multiply two numbers, and these patterns of multiplication are easy to remember. The table shows patterns involved in multiplying numbers by 2, 5, 6, 9, 12, and 20.

PATTERNS OF MULTIPLICATION		
To multiply	How to do it	Example to multiply
2	add the number to itself	$2 \times 11 = 11 + 11 = 22$
5	the last digit of the number follows the pattern 5, 0, 5, 0	5, 10, 15, 20
6	multiplying 6 by any even number gives an answer that ends in the same last digit as the even number	$6 \times 12 = 72$ $6 \times 8 = 48$
9	multiply the number by 10, then subtract the number	$9 \times 7 = 10 \times 7 - 7 = 63$
12	multiply the original number first by 10, then multiply the original number by 2, and then add the two answers	$12 \times 10 = 120$ $12 \times 2 = 24$ $120 + 24 = 144$
20	multiply the number by 10 then multiply the answer by 2	$14 \times 20 =$ $14 \times 10 = 140$ $140 \times 2 = 280$